TASK 5

Write a Pig Latin script to sort, group, join, project, and filter the data.

e.txt

101,John,Sales,65000
102,Alice,HR,70000
103,Bob,IT,80000
104,Eve,Sales,55000
105,Charlie,Finance,90000
106,Mike,IT,75000
107,Susan,HR,72000
108,David,Finance,68000
109,Emma,Sales,62000
110,Robert,IT,78000
111,Olivia,Finance,87000
112,Liam,HR,71000
113,Mia,Marketing,73000
114,Ethan,Marketing,67000
115,Ava,IT,82000

d.txt

Sales,201
HR,202
IT,203
Finance,204
Marketing,205
Operations,206

Run Pig in Interactive Mode: pig -x mapreduce

```
[cloudera@quickstart Desktop]$ pig -x mapreduce
log4j:WARN No appenders could be found for logger (org.apache.hadoop.util.Shell)
.
log4j:WARN Please initialize the log4j system properly.
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more in
fo.
2025-03-19 00:55:29,198 [main] INFO  org.apache.pig.Main - Apache Pig version 0.
12.0-cdh5.13.0 (rexported) compiled Oct 04 2017, 11:09:03
2025-03-19 00:55:29,198 [main] INFO  org.apache.pig.Main - Logging error message
s to: /home/cloudera/Desktop/pig_1742370929177.log
2025-03-19 00:55:29,216 [main] INFO  org.apache.pig.impl.util.Utils - Default bo
otup file /home/cloudera/.pigbootup not found
2025-03-19 00:55:29,709 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
ess
2025-03-19 00:55:29,709 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2025-03-19 00:55:29,709 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.HExecutionEngine - Connecting to hadoop file system at: hdfs://quickstart.clo
```


```
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
ess
2025-03-19 00:55:31,113 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
2025-03-19 00:55:31,113 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
ess
grunt>
```


Load Data:

employees = LOAD 'abc/e.txt' USING PigStorage(',') AS (emp_id:INT, name:CHARARRAY, department:CHARARRAY, salary:INT);

```
grunt> employees = LOAD 'pig/e.txt' USING PigStorage(',') AS (emp_id:INT, name:C
HARARRAY, department:CHARARRAY, salary:INT);
grunt>
```

DUMP employees;

```
cloudera@quickstart:~/Desktop
File  Edit  View  Search  Terminal  Help
ne.util.MapRedUtil - Total input paths to process : 1
(101,John,Sales,65000)
(,,,)
(102,Alice,HR,70000)
(,,,)
(103,Bob,IT,80000)
(,,,)
(104,Eve,Sales,55000)
(,,,)
(105,Charlie,Finance,90000)
(,,,)
(106,Mike,IT,75000)
(,,,)
(107,Susan,HR,72000)
(,,,)
(108,David,Finance,68000)
(,,,)
(109,Emma,Sales,62000)
(,,,)
(110,Robert,IT,78000)
(,,,)
(111,Olivia,Finance,87000)
(,,,)
(112,Liam,HR,71000)
```

departments = LOAD 'abc/d.txt' USING PigStorage(',') AS (department_name:CHARARRAY, manager_id:INT);

```
grunt> departments = LOAD 'abc/d.txt' USING PigStorage(',') AS (department_name:
CHARARRAY, manager_id:INT);
grunt>
```

DUMP departments;

```
2025-03-19 01:02:21,625 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(Sales,201)
(,)
(HR,202)
(,)
(IT,203)
(,)
(Finance,204)
(,)
(Marketing,205)
(,)
(Operations,206)
(,)
(,)
grunt>
```

1. **Filter Employees with Different Conditions**
   1. **Employees earning more than 70000:**
      high_salary = FILTER employees BY salary > 70000;
      DUMP high_salary;

```
2025-03-19 01:08:00,863 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2025-03-19 01:08:00,872 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-19 01:08:00,872 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(103,Bob,IT,80000)
(105,Charlie,Finance,90000)
(106,Mike,IT,75000)
(107,Susan,HR,72000)
(110,Robert,IT,78000)
(111,Olivia,Finance,87000)
(112,Liam,HR,71000)
(113,Mia,Marketing,73000)
(115,Ava,IT,82000)
grunt>
grunt>
```

2. **Employees in the "IT" department:**

   it_employees = FILTER employees BY department == 'IT';

   DUMP it_employees;

```
2025-03-19 01:08:48,224 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2025-03-19 01:08:48,233 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-19 01:08:48,233 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(103,Bob,IT,80000)
(106,Mike,IT,75000)
(110,Robert,IT,78000)
(115,Ava,IT,82000)
grunt>
grunt>
```

3. **Employees earning between 65000 and 80000:**

   mid_salary = FILTER employees BY salary >= 65000 AND salary <= 80000;

   DUMP mid_salary;

```
2025-03-19 01:09:29,812 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2025-03-19 01:09:29,821 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-19 01:09:29,822 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(101,John,Sales,65000)
(102,Alice,HR,70000)
(103,Bob,IT,80000)
(106,Mike,IT,75000)
(107,Susan,HR,72000)
(108,David,Finance,68000)
(110,Robert,IT,78000)
(112,Liam,HR,71000)
(113,Mia,Marketing,73000)
(114,Ethan,Marketing,67000)
grunt>
grunt>
```

2. Project Specific Columns:

1. **Select only employee names and salaries:**

   name_salary = FOREACH employees GENERATE name, salary;

   DUMP name_salary;

```
2025-03-19 01:10:25,309 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(John,65000)
(,)
(Alice,70000)
(,)
(Bob,80000)
(,)
(Eve,55000)
(,)
(Charlie,90000)
(,)
(Mike,75000)
(,)
(Susan,72000)
(,)
(David,68000)
(,)
(Emma,62000)
(,)
(Robert,78000)
(,)
(Olivia,87000)
```

2. **Get employee ID, name, and department (renaming fields):**

   id_name_dept = FOREACH employees GENERATE emp_id AS ID, name AS EmployeeName, department AS Dept;

   DUMP id_name_dept;

```
2025-03-19 01:12:06,188 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(101,John,Sales)
(,,)
(102,Alice,HR)
(,,)
(103,Bob,IT)
(,,)
(104,Eve,Sales)
(,,)
(105,Charlie,Finance)
(,,)
(106,Mike,IT)
(,,)
(107,Susan,HR)
(,,)
(108,David,Finance)
(,,)
(109,Emma,Sales)
(,,)
(110,Robert,IT)
(,,)
```

3. Sorting Employees:

1. **Sort employees by salary (Ascending):**

   sorted_salary_asc = ORDER employees BY salary ASC;

   DUMP sorted_salary_asc;

```
(,,,)
(,,,)
(,,,)
(,,,)
(,,,)
(,,,)
(,,,)
(104,Eve,Sales,55000)
(109,Emma,Sales,62000)
(101,John,Sales,65000)
(114,Ethan,Marketing,67000)
(108,David,Finance,68000)
(102,Alice,HR,70000)
(112,Liam,HR,71000)
(107,Susan,HR,72000)
(113,Mia,Marketing,73000)
(106,Mike,IT,75000)
(110,Robert,IT,78000)
(103,Bob,IT,80000)
(115,Ava,IT,82000)
(111,Olivia,Finance,87000)
(105,Charlie,Finance,90000)
grunt>
grunt>
```

2. **Sort employees by salary (Descending):**

sorted_salary_desc = ORDER employees BY salary DESC;
DUMP sorted_salary_desc;

```
2025-03-20 02:47:25,195 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-20 02:47:25,196 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(105,Charlie,Finance,90000)
(111,Olivia,Finance,87000)
(115,Ava,IT,82000)
(103,Bob,IT,80000)
(110,Robert,IT,78000)
(106,Mike,IT,75000)
(113,Mia,Marketing,73000)
(107,Susan,HR,72000)
(112,Liam,HR,71000)
(102,Alice,HR,70000)
(108,David,Finance,68000)
(114,Ethan,Marketing,67000)
(101,John,Sales,65000)
(109,Emma,Sales,62000)
(104,Eve,Sales,55000)
(,,,)
(,,,)
(,,,)
(,,,)
(,,,)
```

3. **Sort employees by department and then by salary:**

sorted_dept_salary = ORDER employees BY department ASC, salary DESC;
DUMP sorted_dept_salary;

```
(,,,)
(105,Charlie,Finance,90000)
(111,Olivia,Finance,87000)
(108,David,Finance,68000)
(107,Susan,HR,72000)
(112,Liam,HR,71000)
(102,Alice,HR,70000)
(115,Ava,IT,82000)
(103,Bob,IT,80000)
(110,Robert,IT,78000)
(106,Mike,IT,75000)
(113,Mia,Marketing,73000)
(114,Ethan,Marketing,67000)
(101,John,Sales,65000)
(109,Emma,Sales,62000)
(104,Eve,Sales,55000)
grunt>
grunt>
```

4.Group Employees by Department:

grouped_by_dept = GROUP employees BY department;
DUMP grouped_by_dept;

```
2025-03-20 03:14:52,405 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2025-03-20 03:14:52,415 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-20 03:14:52,415 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(HR,{(102,Alice,HR,70000),(112,Liam,HR,71000),(107,Susan,HR,72000)})
(IT,{(110,Robert,IT,78000),(115,Ava,IT,82000),(106,Mike,IT,75000),(103,Bob,IT,80
000)})
(Sales,{(109,Emma,Sales,62000),(101,John,Sales,65000),(104,Eve,Sales,55000)})
(Finance,{(111,Olivia,Finance,87000),(108,David,Finance,68000),(105,Charlie,Fina
nce,90000)})
(Marketing,{(113,Mia,Marketing,73000),(114,Ethan,Marketing,67000)})
(,{(,,,),(,,,),(,,,),(,,,),(,,,),(,,,),(,,,),(,,,),(,,,),(,,,),(,,,),(,,,),(,,,)
,(,,,),(,,,),(,,,)})
grunt>
grunt>
```

5. **Check count of employees in each department:**

dept_count = FOREACH grouped_by_dept GENERATE group AS department, COUNT(employees) AS emp_count;

DUMP dept_count;

```
2025-03-20 03:16:48,956 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2025-03-20 03:16:48,972 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-20 03:16:48,972 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(HR,3)
(IT,4)
(Sales,3)
(Finance,3)
(Marketing,2)
(,0)
grunt>
grunt>
```

6.Join Employees with Department Data:

1. **Simple Join:**

joined_data = JOIN employees BY department, departments BY department_name;

DUMP joined_data;

```
2025-03-20 03:17:37,429 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2025-03-20 03:17:37,447 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-20 03:17:37,448 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(102,Alice,HR,70000,HR,202)
(112,Liam,HR,71000,HR,202)
(107,Susan,HR,72000,HR,202)
(110,Robert,IT,78000,IT,203)
(115,Ava,IT,82000,IT,203)
(106,Mike,IT,75000,IT,203)
(103,Bob,IT,80000,IT,203)
(109,Emma,Sales,62000,Sales,201)
(101,John,Sales,65000,Sales,201)
(104,Eve,Sales,55000,Sales,201)
(111,Olivia,Finance,87000,Finance,204)
(108,David,Finance,68000,Finance,204)
(105,Charlie,Finance,90000,Finance,204)
(113,Mia,Marketing,73000,Marketing,205)
(114,Ethan,Marketing,67000,Marketing,205)
grunt>
grunt>
```

2. **Selecting only specific columns after join:**

joined_selected = FOREACH joined_data GENERATE employees::name, employees::department, departments::manager_id;

DUMP joined_selected;

```
2025-03-20 03:18:25,317 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Ke
y [pig.schematuple] was not set... will not generate code.
2025-03-20 03:18:25,325 [main] INFO  org.apache.hadoop.mapreduce.lib.input.FileI
nputFormat - Total input paths to process : 1
2025-03-20 03:18:25,325 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
(Alice,HR,202)
(Liam,HR,202)
(Susan,HR,202)
(Robert,IT,203)
(Ava,IT,203)
(Mike,IT,203)
(Bob,IT,203)
(Emma,Sales,201)
(John,Sales,201)
(Eve,Sales,201)
(Olivia,Finance,204)
(David,Finance,204)
(Charlie,Finance,204)
(Mia,Marketing,205)
(Ethan,Marketing,205)
grunt>
grunt>
```